2024 SSMO Team Round

SMO Team

TR 1: Find the number of ordered triples of positive integers (a, b, c) that satisfy the equation

$$2(a^b)^c + 1 = 513.$$

TR 2: Find the sum of the three smallest positive integers n such that n^4 leaves a remainder of 1 when divided by 100.

TR 3: Consider four-digit positive integers N such that when N's units digit and thousands digit are removed, what remains is a two-digit perfect square. The average of all such N can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

TR 4: Let ABC be a right triangle with circumcenter O and incenter I such that $\angle ABC = 90^\circ$ and $\frac{AB}{BC} = \frac{3}{4}$. Let D be the projection of O onto AB, and let E be the projection of O onto BC. Let I_1 be the incenter of ADO and let I_2 be the incenter of OEC. The value of $\frac{[I_1 I_2 I]}{[ABC]}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

TR 5: Let ABC be a triangle with $AB = AC = 5$ and $BC = 6$. Let ω_1 be the circumcircle of ABC and let ω_2 be the circle externally tangent to ω_1 and tangent to rays AB and AC. The distance between the centers of ω_1 and ω_2 can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

TR 6: Let α, β, and γ be the roots of the polynomial $x^3 - 6x^2 - 19x - n$. If n is an integer, what is the least possible positive value of $\alpha^3 + \beta^3 + \gamma^3$?

TR 7: Let a and b be real numbers that satisfy

$$a^3 + 8ab^2 = 8b^3 + 4a^2b = 375.$$

Find $\lfloor ab \rfloor$.

TR 8: Let $a < b < c$ be the three integers in $[0, 229)$ that satisfy the congruence $n^3 \equiv 1 \pmod{229}$. Given that $71^2 - 3$ and $107^2 + 1$ are both multiples of 229, find the value of b.

TR 9: Let $ABCDEFGH$ be an equiangular octagon such that $AB = 6$, $BC = 8$, $CD = 10$, $DE = 12$, $EF = 6$, $FG = 8$, $GH = 10$, and $AH = 12$. The radius of the largest circle that fits inside the octagon can be expressed as $a + b\sqrt{c}$, where a, b, and c are positive integers and c is not divisible by the square of any prime. Find $a + b + c$.

TR 10: The side lengths of a convex cyclic quadrilateral $ABCD$ are integers and satisfy

$$(AB \cdot AD + BC \cdot CD)^2 = AC^2 \cdot BD^2 - 72.$$

Find the sum of all possible perimeters of $ABCD$.

TR 11: Let S denote the set of positive divisors of 5400. Let

$$S_i = \{d \mid d \in S,\, d \equiv i \pmod 4\}$$

and let s_i denote the sum of all elements of S_i. Find the value of

$$s_0^2 + s_1^2 + s_2^2 + s_3^2 - 2s_0s_2 - 2s_1s_3.$$

TR 12: What is the smallest positive integer n with exactly three distinct prime factors such that for all integers k, $k^n \equiv k \pmod n$?

TR 13: In a deck of 54 cards consisting of 2 identical jokers and four identical cards of each rank $1, 2, 3, \ldots, 13$, each card is independently dealt to one of 3 people, each having a $\frac{1}{3}$ chance of receiving each card. The expected total, over the three people, of the number of distinct card types each person holds can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

TR 14: Let $a_1, a_2, \ldots, a_7$ be the roots of the polynomial

$$x^7 + 5x^6 + 9x^5 + x^4 + x^3 + 10x^2 + 5x + 1.$$

Find the value of

$$\left| \prod_{n=1}^{7} \prod_{m=n+1}^{7} (a_n a_m - 1) \right|.$$

TR 15: In triangle ABC inscribed in circle ω, let M be the midpoint of BC. Denote P as the second intersection of line AM with ω, where $P \neq A$. If $BP = 9$, $CP = 13$, and $AM = 20$, find the perimeter of triangle ABC.